

Sales Copilot Chatbot: Dynamic RAG Data Endpoint

Architecture, Lifecycle Synchronization & Management UI Implementation Plan

1. Executive Summary & Objective

This implementation plan establishes a first-class **Dynamic Data Endpoint & RAG Synchronization Pipeline** tailored specifically for the **Sales Copilot (AI Chatbot)**. It enables real-time ingestion from external REST APIs, RSS feeds, webhooks, and structured JSON streams directly into the local Hybrid RAG knowledge base (ChromaDB + PostgreSQL).

Core Capabilities:

- **Live Endpoint Registry:** Register and manage external data feeds directly from the Copilot UI.
- **Provenance & Grounding:** Every chunk tracks its originating endpoint URL for strict factual verification.
- **Incremental Hash-Diff Sync:** Compares SHA-256 hashes to embed only modified delta records, saving token quotas.
- **Instant Vector Purge & Zero Stale Data:** Removing an endpoint triggers an immediate cascade deletion across PostgreSQL and ChromaDB, ensuring the chatbot never cites removed sources.

2. System Architecture & Lifecycle Flow

Lifecycle Event	Pipeline Execution Flow	Chatbot Impact
1. Connect / Add Endpoint	API verifies URL -> Fetches payload -> Cleans & Chunks -> Generates dense vectors (FastEmbed bge-small-en-v1.5) -> Writes to ChromaDB & PostgreSQL rag_chunks.	Chatbot immediately discovers new facts and cites them with [n] reference badges.
2. Refresh / Delta Sync	Performs SHA-256 hash comparison. Unchanged documents cost 0 embeddings. Modified records create a new version; old chunks are updated.	Chatbot answers reflect updated numbers/priorities with zero delay.
3. Delete / Remove Endpoint	Cascade Purge triggers: Removes all matching vector IDs from ChromaDB and deletes records from rag_documents and rag_chunks.	Chatbot immediately stops citing the deleted endpoint, preventing zombie data & hallucinations.

3. Technical Implementation Deliverables

A. Database Schema Additions (`db/models` & PostgreSQL)

- Table rag_endpoints: id, name, url, endpoint_type, account_id, is_active, last_synced_at, document_count, chunk_count.
- Provenance Column: rag_documents.endpoint_id with ON DELETE CASCADE for automated cleanup.

B. Endpoint Lifecycle API (`apps/sales\_copilot/api.py`, `endpoints.py`)

- POST /api/copilot/endpoints: Register and validate a new data endpoint.
- GET /api/copilot/endpoints: List all active endpoints, health status, and chunk metrics.
- POST /api/copilot/endpoints/{id}/sync: Trigger an on-demand hash-diff synchronization.
- DELETE /api/copilot/endpoints/{id}: Disconnect endpoint and purge ChromaDB vectors + PostgreSQL chunks.

C. Ingestion, Chunking & Vector Purge Engine (`apps/sales\_copilot/ingest.py`, `store.py`)

- Ingests JSON arrays, REST payloads, RSS XML, and clean text.
- Embeds semantic chunks locally using fastembed without external API cost.
- Synchronous vector invalidation: removes vectors matching endpoint_id from ChromaDB on deletion.

D. Sales Copilot UI Console (`frontend/templates/copilot.html`, `endpoints-ui.js`)

- Dedicated **Data Sources & Endpoints** management modal inside the Copilot workspace.
- Live status indicators (*Active*, *Syncing*, *Error*) and document/chunk counters.
- Quick **Add Endpoint** modal with live URL connection test.
- One-click **Sync Now** and **Remove** actions with progress toasts and confirmation modals.

4. Verification & Testing Protocol

1. **Ingestion Test:** Add a test endpoint with corporate news/priorities and trigger sync. Verify vectors appear in ChromaDB and chunks in PostgreSQL.
2. **Chatbot Grounding Test:** Query the chatbot about specific facts from the endpoint. Verify accurate answer and citation [1] referencing the endpoint URL.
3. **Purge & Invalidation Test:** Delete the endpoint from the UI. Confirm vectors and rows are purged. Re-ask the question; verify the chatbot reports no data is found (0% hallucination).